

Homework

1. Show that $\mathbb{Z}_6$ is a ring with unity under addition and multiplication; this covers parts

Proof. We know that $\mathbb{Z}_6$ is a commutative group with additive identity 0. This covers parts 1-5 of the definition of a ring. Using the Cayley tables¹:

$\times$	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	0	2	4
3	0	3	0	3	0	3
4	0	4	2	0	4	2
5	0	5	4	3	2	1

Table 1: Cayley tables for $\times$ in $\mathbb{Z}_6$.

From the table we can see that $\mathbb{Z}_6$ is closed under multiplication. Association and distribution is preserved in modular arithmetic so $\mathbb{Z}_6$ is a ring. $\square$

- Note that we can also see from the table that $\mathbb{Z}_6$ has a multiplicative identity so it is a ring with unity, but every element does not have a multiplicative inverse.

2. Show that the set $\sqrt{2}\mathbb{Z} = \{x: x = m + \sqrt{2}n \text{ for some } m, n \in \mathbb{Z}\}$ is a ring.

Proof. Again we know that $\sqrt{2}\mathbb{Z}$ is a commutative group under addition. So we only need to show that parts 6-8 of the definition hold.

(part 6): Let $a, b \in \sqrt{2}\mathbb{Z}$ then $a = m_1 + n_1\sqrt{2}$ and $b = m_2 + n_2\sqrt{2}$ for some $m_1, m_2, n_1, n_2 \in \mathbb{Z}$.

$$\begin{aligned}
 ab &= (m_1 + \sqrt{2}n_1)(m_2 + \sqrt{2}n_2) \\
 &= m_1m_2 + \sqrt{2}n_1m_2 + \sqrt{2}n_2m_1 + 2n_1n_2 \\
 &= (2n_1n_2 + m_1m_2) + \sqrt{2}(m_1n_2 + m_2n_1)
 \end{aligned}$$

Since integers are closed under addition, $(2n_1n_2 + m_1m_2)$ and $(m_1n_2 + m_2n_1)$ are both integers $ab \in \sqrt{2}\mathbb{Z}$.

Parts 7 and 8 hold since any number in $\sqrt{2}\mathbb{Z}$ is also a real number, and distribution and association hold in $\mathbb{R}$. So $\sqrt{2}\mathbb{Z}$ is a ring. $\square$

3. Let F be the set of all functions from $\mathbb{R}$ to $\mathbb{R}$. Define addition and multiplication as

$$(f + g)(x) = f(x) + g(x) \text{ and } (fg)(x) = f(x)g(x).$$

Show that F is a commutative ring with identity.

Proof. Parts 1 and 6. Let $f, g \in F$. Because f and g are both functions, each x is mapped to two and only two real numbers: $f(x)$ and $g(x)$ (f and g are well defined). The operations of addition and multiplication are also well defined so then $f(x) + g(x)$ and $f(x)g(x)$ each map x to one real number. So then $(f + g)(x)$ and $(fg)(x)$ are functions in F .

Part 8. Let $f, g, h \in F$. $(h(f + g))(x) = h(x)(f + g)(x) = h(x)(f(x) + g(x))$. So distribution holds.

Parts 2, 3, and 7 hold because they hold in $\mathbb{R}$.

Parts 4 and 5. 0_F is the zero function $f(x) = 0$. $g(x) = -1$ is also a function over $\mathbb{R}$; so then the additive inverse of $f(x)$ is just $-1f(x)$.

Finally $f(x) = 1$ is the multiplicative inverse. So F is a ring with identity. $\square$

¹A Cayley table shows all the possible products/sums of all a ring's elements.

4. Prove that the ring above, F , is not an integral domain.

Hint Here are two *indicator functions*

$$\mathbf{1}_{[0,\infty)}(x) = \begin{cases} 1 & \text{if } x \in [0, \infty) \\ 0 & \text{if } x \notin [0, \infty) \end{cases} \text{ and } \mathbf{1}_{(-\infty,0]}(x) = \begin{cases} 1 & \text{if } x \in (-\infty, 0] \\ 0 & \text{if } x \notin (-\infty, 0] \end{cases}$$

The indicator function is a function that has an output of either 1 or 0. For example, $\mathbf{1}_{[0,\infty)}(2) = 1$ and $\mathbf{1}_{[0,\infty)}(-2) = 0$. Now consider the functions $f(x) = x\mathbf{1}_{[0,\infty)}$ and $g(x) = x\mathbf{1}_{(-\infty,0]}$.

Proof. Following the hint, $f(x) \neq 0_F$ and $g(x) \neq 0_F$ but $f(x)g(x) = x\mathbf{1}_{[0,\infty)}x\mathbf{1}_{(-\infty,0]} = x^2\mathbf{1}_{[0,\infty)}\mathbf{1}_{(-\infty,0]} = 0$ if $x = 0$ then $f(0)g(0) = 0 \cdot 1^2 = 0$ and if $x \neq 0$ then $f(x)g(x) = x^2 \cdot 0 = 0$. So F is not an integral domain. $\square$

5. Let R and S be rings. Define addition and multiplication on the Cartesian product $R \times S$ by,

$$(r, s) + (r', s') = (r + r', s + s') \text{ and } (r, s)(r', s') = (rr', ss')$$

for $r, r' \in R$ and $s, s' \in S$. Prove that $R \times S$ is a ring.

Proof. Since R and S are rings $r + r', rr' \in R$ and $s + s', ss' \in S$ so then $(r + r', s + s'), (rr', ss') \in R \times S$. Similarly, parts 2, 3, 5, and 7 hold.

$(r'', s'')((r, s) + (r', s')) = (r''(r + r'), s''(s + s'))$ for element in R and S so then distribution holds.

Finally, $0_{R \times S} = (0_R, 0_S)$ so $R \times S$ is a ring. $\square$

6. Show that the set $k\mathbb{Z} = \{x: x = kn \text{ for some } n \in \mathbb{Z}\}$ for any integer k is a subring of $\mathbb{Z}$.

Proof. Let $n, m \in \mathbb{Z}$ so that $kn, km \in k\mathbb{Z}$. $(kn)(km) = k(knm) \in k\mathbb{Z}$, and $kn - km = k(n - m) \in k\mathbb{Z}$ since $kn, n - m \in \mathbb{Z}$. We know that $\mathbb{Z}$ is a ring. So then $k\mathbb{Z}$ is a subring (and also a ring) by Theorem 1.3 (the subring test). $\square$

7. Prove that $\mathbb{Q}$ is an integral domain.

Proof. Let $a, b, c, d \in \mathbb{Z} - \{0\}$ so that $\frac{a}{b}, \frac{c}{d}$ are nonzero elements in $\mathbb{Q}$. Then $\frac{a}{b} \frac{c}{d} = \frac{ac}{bd} \neq 0$ since $ac \neq 0$ since $\mathbb{Z}$ is an integral domain. $\square$

8. Prove that $\mathbb{Z}_6$ is not an integral domain.

Proof. Consider the nonzero elements $2, 3 \in \mathbb{Z}_6$, and $2 \cdot 3 = 0 \pmod{6}$. Also see table 1. $\square$

9. Let a and b be two nonzero elements in a ring R . Prove that in general, $ax = b$ does not have a unique solution.

Hint It should be understood that x must be in R . An example is all that is needed. You can find an example that has two solutions, or find an example with no solution.

Proof. $\mathbb{Z}_6$ is a ring, but $2 \cdot 3 = 0 \pmod{6}$ and $2 \cdot 0 = 0 \pmod{6}$. So the equation $2x = 0$ has more than one solution. See table 1 for more examples. $\square$

10. Prove that $\mathbb{Z}_n$ is not an integral domain (under the usual addition and multiplication) if n is not a prime (n is assumed to be a positive integer).

Hint If n is not prime then $n = mq$ for two integers $1 < m, q < n$. Looking at the example seen in exercise might help.

Proof. Since n is not prime we can write $n = mq$ for two integers $1 < m, q < n$. $1 < m < n$ and $1 < q < n$ so both are nonzero elements in $\mathbb{Z}_n$, and $mq = n = 0 \pmod n$. So $\mathbb{Z}_n$ is not an integral domain. $\square$

Presentation Problems

1. Prove that the intersection of any two subrings of a ring R is also a subring of R .

Proof. Let S, T be two subrings of R . $S \cap T$ is a nonempty subset of R since $0_R \in S$ and $0_R \in U$. Let $a, b \in S \cap T$. $ab \in S$ and $a - b \in S$ since S is a ring. Similarly $ab \in T$ and $a - b \in T$. So $ab, a - b \in S \cap T$, and $S \cap T$ is a subring of R by Theorem 1.3. $\square$

2. Give an example of a ring with elements a and b with properties that $ab = 0$ but $ba \neq 0$.

Proof. Let $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \in M_{2 \times 2}(\mathbb{R})$. The zero in this ring is $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} \text{ and } \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \square$$

3. Let m and n be positive integers with least common multiple k . Prove that $m\mathbb{Z} \cap n\mathbb{Z} = k\mathbb{Z}$.

Proof. Since $k = \text{lcm}(n, m)$, $k = na = mb$ for some $a, b \in \mathbb{Z}$. So then $xk = x(na) = x(mb)$ for any integer x and $xk \in m\mathbb{Z} \cap n\mathbb{Z}$. Thus $m\mathbb{Z} \cap n\mathbb{Z} \supseteq k\mathbb{Z}$.

Let $x \in m\mathbb{Z} \cap n\mathbb{Z}$, so then $x = ma = nb$ meaning x is a common multiple of m and n . Since k is the least common multiple, $k \leq x$ and $k|x$.² So then $m\mathbb{Z} \cap n\mathbb{Z} \subseteq k\mathbb{Z}$. $\square$

4. Prove that a ring that is cyclic under addition is commutative.

Proof. Recall that cyclic under addition means that there is a element a in a ring R such that for any $r \in R$ $a + a + \cdots + a = r$. Let $s, t \in R$. $st = \underbrace{(a + a + \cdots + a)}_m \underbrace{(a + a + \cdots + a)}_n = mna^2 = \underbrace{(a + a + \cdots + a)}_n \underbrace{(a + a + \cdots + a)}_m = ts$. $\square$

5. Prove that if a ring element has a multiplicative inverse, it is unique.
6. Prove that a field is an integral domain.
7. Prove that $\mathbb{Z}_p$ is an integral domain if p is prime (you can assume that it is a commutative ring with identity).
8. Prove that $\mathbb{Z}_p$ is a field.

²This was shown in MTH 335.